

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points. When writing proofs use COMPLETE SENTENCES! I will take away points otherwise.

Problem 1

Using the Archimedean Property, give a direct $\epsilon - N$ proof that

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0.$$

Problem 2

Using the Archimedean Property, give a direct $\epsilon - N$ proof that the following sequence converges:

$$\left\{ \frac{2}{\sqrt{n}} + \frac{1}{n} + 3 \right\}$$

Problem 3

Prove that the Archimedean Property is equivalent to the fact that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, that is show that the two imply each other.

Problem 4

For each of the following statements, determine whether it is true or false and justify your answer.

- a) Every bounded sequence converges.
- b) A convergent sequence of positive numbers has a positive limit.
- c) The limit of a convergent sequence is the interval (a, b) also belongs to (a, b) .

Problem 5

Let $\{a_n\}$ be a monotone sequence. Prove that $\{a_n\}$ converges if and only if $\{a_n^2\}$ converges.

Problem 6

For each of the following statements, determine whether it is true or false and justify your answer.

- a) A sub-sequence of a bounded sequence is bounded.
- b) A sub-sequence of a monotone sequence is monotone.
- c) A sub-sequence of a convergent sequence is convergent.
- d) A sequence converges if it has a convergent sub-sequence.

Problem 7

For each of the following sequences, find the peak indices (NOT VALUES).

a) $\left\{\frac{1}{n}\right\}$

b) $\{(-1)^n\}$

c) $\left\{\frac{(-1)^n}{n}\right\}$

Problem 8

Prove that a sequence $\{a_n\}$ is bounded if and only if there are numbers a and b with $a < b$ such that $\{a_n\}$ is a sequence in the interval $[a, b]$.

Extra Credit:

Define the sequence $\{s_n\}$ by $s_n = \frac{1}{2 \cdot 1} + \frac{1}{3 \cdot 2} + \dots + \frac{1}{(n+1)(n)}$ for all $n \in \mathbb{N}$. Prove that $\lim_{n \rightarrow \infty} \{s_n\} = 1$.